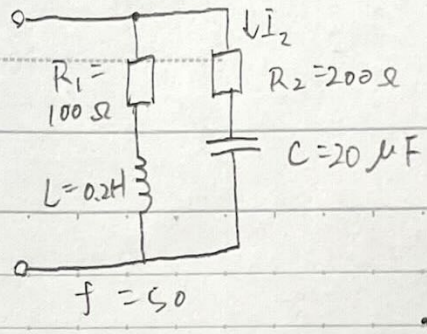
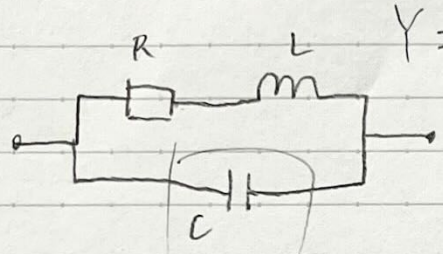



$\dot{Z}_L = 100 + 314 \times 0.2\,i$

$\dot{Z}_C = 200 - \dfrac{1}{314 \times 20 \times 10^{-6}}\,i$

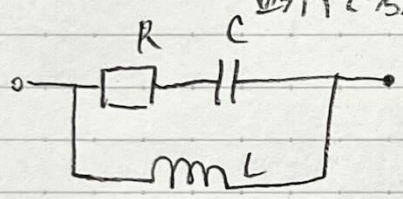



抵抗のない
並列Yで考える.

$$\dot{Y} = \frac{1}{R + j\omega L} + j\omega C$$

$$= \frac{R - j\omega L}{(R + j\omega L)(R - j\omega L)} + j\omega C$$

$$= \frac{R - j\omega L}{R^2 + \omega^2 L^2} + j\omega C$$




$$\dot{Y} = \frac{1}{R - j\frac{1}{\omega C}} - j\frac{1}{\omega L}$$

$$= \frac{\omega C}{R\omega C - j}$$

$$= \frac{\omega C(R\omega C + i)}{R^2\omega^2 C^2} - j\frac{1}{\omega L}$$

皮相電力

$P_a = IV$ [VA]　　$P = IV\cos\theta$ [W]

$$\theta = (\theta_I - \theta_V)$$

例15.3 インピーダンス $\dot{Z} = R + jX = Z\angle\theta_Z$ , $\dot{V} = V\angle\theta_V$

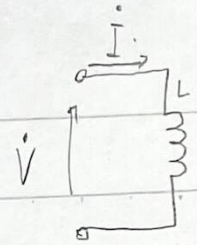
$\dot{Z} = \dfrac{\dot{V}}{\dot{I}}$

$$\dot{I} = \frac{\dot{V}}{\dot{Z}} = \frac{V\angle\theta_V}{Z\angle\theta_Z} = \frac{V}{Z}(\theta_V - \theta_Z) = I\angle\theta_I$$

$I = \dfrac{V}{Z}$ , $\theta_I = \theta_V - \theta_Z$

$\theta = \theta_I - \theta_V = -\theta_Z$

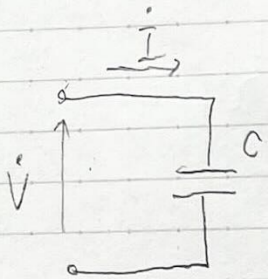
$P = IV\cos\theta = \dfrac{V^2}{Z}\cos\theta_Z$




$\dot{V} = \omega L \angle 90^\circ \times \dot{I}$

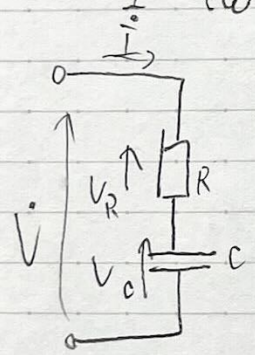
$\dot{V} = j\omega L \dot{I}$ , $\dot{I} = -\frac{\dot{V}}{\omega L} j$

$I_L$




$\dot{I} = \omega C \angle 90^\circ \times \dot{V}$

$\dot{I} = j\omega C \dot{V}$ , $\dot{V} = -j \frac{\dot{I}}{\omega C}$




$\dot{V} = \dot{V}_R + \dot{V}_C$

$= \left(R - j\frac{1}{\omega C}\right)\dot{I}$

$\dot{V}_C = -j\frac{\dot{I}}{\omega C}$ $\dot{I} =$

インピーダンス

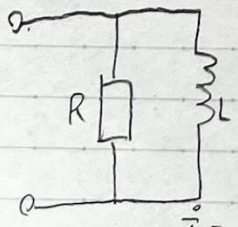
R-L直列 $\dot{Z} = R + j\omega L$

R-C直列 $\dot{Z} = R - j\frac{1}{\omega C}$

アドミタンス

R-L並列 $\dot{Y} = \frac{1}{R} - j\frac{1}{\omega L}$

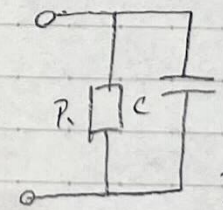
R-C並列 $\dot{Y} = \frac{1}{R} + j\omega C$




$\dot{I}_L = -j\frac{\dot{V}}{\omega L}$

$\dot{I}_R = \frac{\dot{V}}{R}$

$\dot{I} = \left(\frac{1}{R} - j\frac{1}{\omega L}\right)\dot{V}$




$\dot{I}_R = \frac{\dot{V}}{R}$

$\dot{I}_C = j\omega C \dot{V}$

$\dot{I} = \left(\frac{1}{R} + j\omega C\right)\dot{V}$